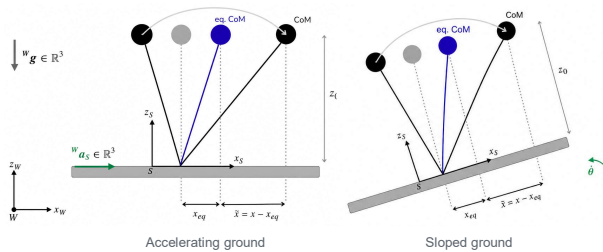


Overview: Sloped and accelerating ground alters the reduced-order dynamics underlying humanoid walking. B-LIP accounts for these effects through an effective-gravity-induced equilibrium shift while preserving the nominal LIP structure, enabling use with any LIP-guided controller. We demonstrate B-LIP using an H-LIP deadbeat step planner, comparing it against the baseline H-LIP planner and an adaptive ankle-torque controller.

BACKGROUND

LIP provides a direct CoM-foot-placement relation that underlies capture-point, DCM, H-LIP, and LIP-guided learning methods [2–4,7]. On moving ground, prior work changes gait generation / foot placement when motion is known [10], uses robust or optimization-based control for bounded motion [11], or trains policies over randomized platform motion [12]. For unknown platform motion, adaptive ankle torque can compensate online [13], but ankle torque and feasible step length are hardware-limited. **Our change is at the model layer:** shift the LIP equilibrium so the robot adapts to the altered dynamics instead of rejecting them.

B-LIP: ONE MODEL FOR SLOPE + ACCELERATING GROUND



Both effects alter the effective acceleration in the surface frame. Their different physics collapse into the same shifted equilibrium.

$$s g_{\text{eff}} = s R_W ({}^W g - {}^W a_s)$$

$$\ddot{x} = \frac{g_{\text{eff},z}}{z_0} x - g_{\text{eff},x} \quad x_{\text{eq}} = z_0 \frac{g_{\text{eff},x}}{g_{\text{eff},z}}$$

$$\ddot{x} = x - x_{\text{eq}}, \quad \alpha = \sqrt{\frac{g_{\text{eff},z}}{z_0}}$$

$$\ddot{x} = \alpha^2 \tilde{x}$$

Key property: after translating by x_{eq} , the dynamics recover nominal LIP. Therefore, existing LIP-guided controllers can operate on the shifted state rather than requiring external compensation or model redesigning.

STABILITY STRUCTURE RETAINED

A constant translation preserves the nominal LIP phase-space structure and closed-loop eigenvalues.

$$E(\tilde{x}, v) = v^2 - \alpha^2 \tilde{x}^2, \quad \dot{E} = 0$$

$$P_B = \Phi^{-1} \circ P_0 \circ \Phi$$

WHY THIS IS GENERALIZABLE ACROSS CONTROLLERS

The modification is a coordinate / model update, not a controller-specific stabilization trick.

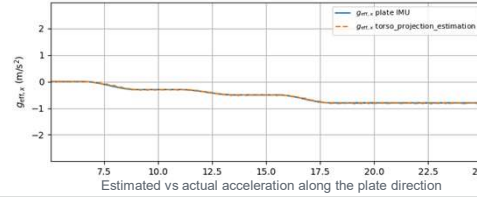


H-LIP IMPLEMENTATION

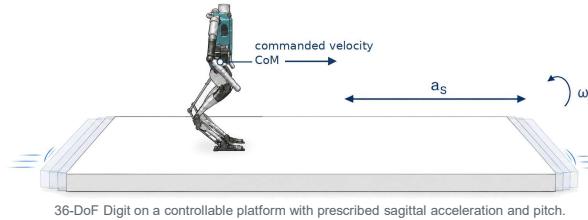
We deliberately chose H-LIP deadbeat step-planner as our main implementation for evaluation of the idea. This is because the step-planner directly inherits LIP structure and uses the original non-robust deadbeat controller [4,6] that force recovery per step. Keeping the controller unchanged while **modifying only x_{eq} and α** isolates the benefit of B-LIP model adaptation.

ONBOARD EFFECTIVE-GRAVITY ESTIMATION

To apply B-LIP **using onboard sensors**, we estimate ground motion from the torso IMU and differentiated stance-foot kinematics. Subtracting foot acceleration relative to the torso yields surface acceleration; we express gravity minus this acceleration in the surface frame derived from foot kinematics.



EVALUATION SETUP



Evaluation setup: 36-DoF Digit in MuJoCo on a platform with prescribed sagittal acceleration and pitch. B-LIP updates the H-LIP reference online using estimated effective gravity; the resulting CoM and swing-foot references are mapped to the full-order robot through the same constrained IK and passivity-based low-level controller used across all comparisons.

COMPARISON

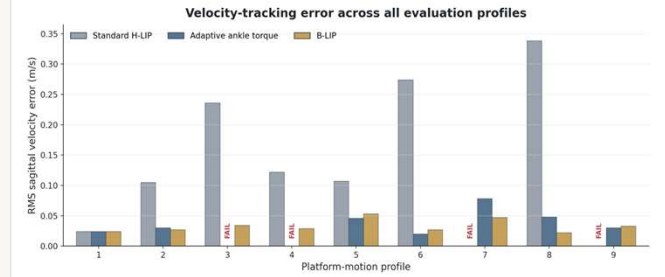
1 Stationary $a_s = 0$

TRANSLATION	PITCH
2 $a_s: 0 \rightarrow -0.2 \rightarrow -0.4 \text{ m/s}^2$	6 $\theta: 0^\circ \rightarrow 3^\circ \rightarrow 6^\circ$
3 $a_s: 0 \rightarrow 0.4 \rightarrow 0.8 \text{ m/s}^2$	7 $\theta: 0^\circ \rightarrow -9^\circ \rightarrow -17^\circ$
4 $a_s: 0 \rightarrow 0.5 \rightarrow 0 \text{ m/s}^2$	8 $\theta: 0^\circ \rightarrow -8^\circ \rightarrow 0^\circ$
5 $x_s = 0.25[1 - \cos(2\pi(t-2)/5)] \text{ m}$	9 $\theta = 5^\circ [1 - \cos(0.4\pi t)]$

Piecewise: 6.5 s constant intervals + 2.5 s ramps. Cases 5 and 9 are continuous.

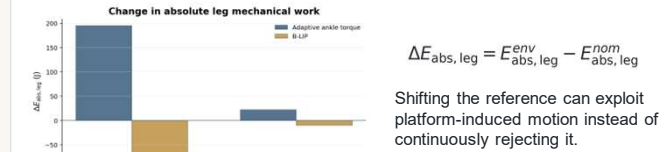
The above profiles were evaluated with commanded walking velocity of 0.2 m/s

TRACKING + ROBUSTNESS RESULTS



84% lower mean RMS error vs. H-LIP and 17% lower mean RMS error vs. adaptive ankle torque on the shared successful nonstationary profiles.

MECHANICAL WORK: ADAPT RATHER THAN FIGHT



TAKEAWAY + LIMITATIONS

Takeaway: Accounting for slope- and acceleration-induced effects directly in the LIP model improves velocity tracking and robustness without redesigning the underlying controller.

Limitations: Current evaluation is simulation-only, sagittal, and limited to low commanded velocity. The estimator is also susceptible to sensor noise and latency, model mismatch, and real-time implementation constraints, which may degrade estimation accuracy.